

Na Min An

PH.D. STUDENT

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Research Interests

My research aims to develop assistive and trustworthy AI and machine learning systems grounded in an understanding of human perception, cognition, and action. I am particularly interested in identifying and mitigating modality-specific and social biases in vision-language models, using a mechanistic interpretability framework to uncover the internal representations and decision-making processes that give rise to biased behaviors.

Education

Korea Advanced Institute of Science and Technology (KAIST)

Seoul, South Korea

DOCTOR OF PHILOSOPHY IN ARTIFICIAL INTELLIGENCE, GPA: 4.13/4.3

Feb. 2028 (Expected)

- Advisors: Prof. Hyunjung Shim (2024-Current) and Dr. James Thorne (2023-2025)

University of Seoul

Seoul, South Korea

BACHELOR OF SCIENCE IN MATHEMATICS, GPA: 4.29/4.5

Feb. 2022

- Summa Cum Laude (Received academic excellence or Mayor's scholarship for every semester)

Skills

Language Python, MATLAB, R, C++, JavaScript

Certification Society of Actuaries (P, FM, and SRM)

Experience

KAIST

Seoul, South Korea

RESEARCH ASSISTANT

March 2023 - PRESENT

- Nominated for Apple PhD Fellowship program (3 students per institution)
- Ministry of Science and ICT Project: Contributed leadership to a national project aimed at developing socially unbiased generative models.
- Basic Research Lab Project: Co-led a national project on developing barrier-free AI systems in educational and mobility domains.
- Published six first-author and three co-authored Top-tier works, with research centered on fairness and human-centered ML systems.

University of Copenhagen

Copenhagen, Denmark

VISITING RESEARCHER

March 2025 - May 2025

- Received a Danish Data Science Academy grant to advance research on explainability and reliability in multimodal models to build fairness-aware ML systems.
- Developed a neuron-level bias mitigation framework for VLMs and LVLMs with Dr. Yoonna Jang, in collaboration with the NVIDIA Taiwan team (Dr. Yusuke Hirota and Dr. Ryo Hachiuma).

Korea Institute of Science and Technology (KIST)

Seoul, South Korea

RESEARCH INTERN

Oct. 2022 - Feb. 2023

- Analyzed human perceptual limits and ML model performance under severely low-resolution visual conditions.
- Developed a reinforcement learning-based framework that emulates the learning effects frequently occurring in human psychophysical tests.
- Built an automated imaging pipeline for detecting fluorescence-image edges and identifying hyperpolarization phenomena with high precision.

University of Seoul

Seoul, South Korea

RESEARCH INTERN

Sep. 2020 - Dec. 2020

- Investigated district-level Covid-19 risk factors across Seoul and awarded 3rd place (out of 40+ teams) in a competitive big-data competition.
- Developed a hybrid CNN-RNN model to classify the progressive stages of pulmonary nodules from sequential CT scan slices.
- Analyzed YouTube channel performance to model subscriber trajectory and provided targeted strategies for subscriber acquisition.

Working Projects

Large Vision-Language Models for Blind and Low-Vision Individuals

- Built metrics to improve BLV-model evaluation scoring alignment (VLM: [arXiv:2502.14883](#), LVLM: [arXiv:2510.00766](#)).
- Planning to develop a lightweight, real-time LVLM evaluator for deployment in assistive applications for BLV users.

Multilingual Referring Image Segmentation

- Developed a training-free framework that leverages untapped spatial knowledge in VLMs to enhance zero-shot RIS ([arXiv:2509.23098](#)).
- Extending the framework to a multilingual VLM encoder to enable zero-shot RIS across diverse languages and linguistic structures.

Selected Publications

1. **An, N. M.**, Jang Y., Hirota, Y., Hachiuma, R., Augenstein, I., and Shim, H., “Interpretable Debiasing of Vision-Language Models for Social Fairness,” [CVPR](#), 2026.
2. Kim, E.*, Park, J.*, **An, N. M.***, Kim, J., Patel, H. L., Jin, J., Kruk, J., Agarwal, A., Panda, S., Ilasariya, F. A., Shim, H., and Oh, A., “World in a Frame: Understanding Culture Mixing as a New Challenge for Vision-Language Models,” [CVPR](#) (* denotes equal contribution), 2026.
3. Waheed, S.* and **An, N. M.***, “Image Embedding Sampling Method for Diverse Captioning,” [EMNLP Main](#) (* denotes equal contribution), 2025.
4. **An, N. M.***, Kim, E.*, Thorne, J., and Shim, H., “IoT: Embedding Standardization Method Towards Zero Modality Gap,” [ACL Outstanding](#) (Top 1%, 26 out of 3,000 accepted papers, * denotes equal contribution), 2025.
5. Bayramli, Z.*, Suleymanzade, A.*, **An, N. M.**, Ahmad, H., Kim, E., Kang, J., Thorne, J., and Oh, A., “Diffusion Models Through a Global Lens: Are They Culturally Inclusive?” [ACL Oral](#) (Top 8%, 243 out of 3,000 accepted papers, * denotes equal contribution), 2025.
6. Kang, W. J., Kim, E., **An, N. M.**, Kim, S., Choi, H., Kwak, K., and Thorne, J., “Sightation Counts: Leveraging Sighted User Feedback in Building a BLV-aligned Dataset of Diagram Descriptions,” [ACL Main](#), 2025.
7. **An, N. M.**, Roh, H., Kim, S., Kim, J. H., and Im, M., “Machine Learning Techniques for Simulating Human Psychophysical Testing of Low-Resolution Phosphene Face Images in Artificial Vision,” [Advanced Science](#) (Impact Factor: 14.3, JCR Ranking: 6.5%), 2025 - Featured in [Korean National News](#).
8. Jung, W., Hong, J., **An, N. M.**, Thorne, J., and Yun, S., “Stable Language Model Pre-training by Reducing Embedding Variability,” [EMNLP Main](#), 2024.
9. Lee, N.*, **An, N. M.***, and Thorne, J. “Can Large Language Models Infer and Disagree Like Humans?” [EMNLP Main](#) (* denotes equal contribution), 2023.

Publications

1. Kim, E.*, Park, J.*, Park, N.*, **An, N. M.***, Kim, K., Yoon S., Huo, J., and, Shim, H., “Aligned but Stereotypical? The Hidden Influence of System Prompts on Social Bias in LVLM-Based Text-to-Image Models,” [ICLR RSI Workshop](#) (* denotes equal contribution), 2026.
2. Waheed, S., **An, N. M.**, Milford, M., Ramchurn, S. D., and Ehsan, S., “VLM-Guided Visual Place Recognition for Planet-Scale Geo-Localization,” [ACRA](#), 2025.
3. Kim, J. S*, Yu, A*, Ismayilzada J*, Park, J., Kim, E., Ahmad, H., **An, N. M.**, Thorne, J., and Oh, A., “WHEN TOM EATS KIMCHI: Evaluating Cultural Awareness of Multimodal Large Language Models in Cultural Mixture Contexts,” [NAACL C3NLP Workshop Outstanding](#) (* denotes equal contribution), 2025.
4. **An, N. M.**, Weed, S., and Thorne, J., “Capturing the Relationship Between Sentence Triplets for LLM and Human-Generated Texts to Enhance Sentence Embeddings,” [EACL Findings](#), 2024.
5. **An, N. M.**, Roh, H., Kim, S., Kim, J. H., and Im, M. “Reinforcement Learning Framework to Simulate Short-Term Learning Effects of Human Psychophysical Experiments Assessing the Quality of Artificial Vision,” [IJCNN Oral](#), 2023.

6. **An, N. M.**, Roh, H., Jung, S., Kim, E. J., and Im, M. "Machine Learning Can Approaches as an Alternative to Human Psychophysical Tests of Prosthetic Vision," *EMBS Abstract*, 2021.

Patents

1. Im, M., Roh, H., and **An, N. M.**, "Artificial Vision Parameter Learning and Automating Method for Improving Visual Prosthetic Systems", *US Patent Granted No. F06639*, 2026.
2. Im, M., Roh, H., and **An, N. M.**, "Artificial Vision Expression Parameter Automation Learning System Method for Improving Artificial Vision Device", *Korea Paper Granted No. 0172619*, 2021.

Service

Talks

- AI Tutorial Series at KAIST AI (Winter 2025).

Volunteer

- Provided on-site support for EMNLP conference (2023).
- Contributed to community support and educational outreach in Cambodia in Summer 2018.

Leader

- Lab representative student from March 2024 to Aug. 2025.
- Appeared in the KAIST AI promotional video (2024) and top-10 research achievements participant (2023).
- Lab research manager from Aug. 2023 to Dec. 2023.

Teaching Assistant

- Department Seminars & Colloquia, Fall 2023 & Spring 2026
- Computational Image Generation and Modification, Fall 2025

Mentor

- Led research mentorship for two Azerbaijan undergraduate students in South Korea from Sep. 2024 to Jan. 2025.
- Provided English and mathematics tutoring to Dongdaemun-gu (Seoul, South Korea) high school students in Fall 2019 and Winter 2020.

Reviewer

- IEEE Transactions on Audio, Speech and Language Processing 2025 and 2026
- ICLR RSI Workshop 2026
- ACL ARR February, May, July, October 2025, January 2026
- ACL Student Research Workshop 2025 and 2026
- IJCNN 2024, 2025, and 2026

References

Hyunjung Shim

ASSOCIATE PROFESSOR, KAIST AI

- Email: kateshim@kaist.ac.kr

March 2024 - PRESENT

Seoul, South Korea

James Thorne

CHIEF AI SCIENTIST, THEIA INSIGHTS (PAST: ASSISTANT PROFESSOR, KAIST AI)

- Email: james@jamesthorne.com

March 2023 - Aug. 2025

Cambridge, United Kingdom

Yoonna Jang

POST-DOC, UNIVERSITY OF COPENHAGEN

- Email: morelychee@gmail.com

March. 2025 - Dec. 2025

Copenhagen, Denmark

Isabelle Augenstein

PROFESSOR, DEPARTMENT OF COMPUTER SCIENCE, UNIVERSITY OF COPENHAGEN

- Email: augenstein@di.ku.dk

March 2025 - Dec. 2025

Copenhagen, Denmark

Alice Oh

PROFESSOR, KAIST SCHOOL OF COMPUTING

- Email: alice.oh@kaist.edu

Sep. 2024 - Dec. 2025

Daejeon, South Korea

Maesoon Im

PRINCIPAL RESEARCH SCIENTIST, BRAIN SCIENCE INSTITUTE, KIST

- Email: maesoon.im@kist.re.kr

Oct. 2021 - Feb. 2023

Seoul, South Korea